

Multifunction Human Detection And WEATHER MONITORING SYSTEM

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A human detection system was designed earlier to utilise the microwave Doppler effect to detect human movement and trigger AC appliances such as light bulbs and other electrical loads up to 1A. The design incorporated a

solder jumper pad for mode selection, enabling the same PCB to be configured for IoT compatibility by integrating IoT functionality into the radar switch. This configuration supports real-time radar data monitoring, alert reception, and

wireless control of lights over a Wi-Fi network.

Coverage of the IoT code and app development for this system was planned in the previous month. Building on the existing architecture, the software and circuitry required



Fig. 1: Radar UI dashboard and radar PCB design

BILL OF MATERIALS

Name	Designator	Footprint	Quantity	Manuf. Part	Manufacturer	Indian Substitute	Indian Manufacturer
RCWL-0516 microwave radar module	U1	RCWL-0516	1	RCWL-0516		NA	NA
LDR	U2	UNIVIBE LAMP-LDR ARRANGEMENT_LDR3	1			NA	NA
IndusBoard	U3	FPC24 0.5mm	1	Coin V2	NA	IndusBoard Coinv2	NA
100nF	C3	C1206	1	CC1206KKX-7RYBB104	YAGEO	1206B104K251CT	Walsin
DB301V-3.5-2P-GN	CON1, CON2	CONN-TH_DB301V-3.5-2P-GN	2	DB301V-3.5-2P-GN-S	DORABO	TL001R-3.5-2P	Connectwell
FPC Connector	FPC1	OV2640 FPC 24P	1	AFC01-S24FCA-00	JUSHUO	ZF5S-24-01-T-WT	Samtec Electronics India
VO2223A X007T	IC1	DIP-8_7P-L9.7-W6.4-P2.54-LS7.6-BL-PE7	1	VO2223A-X007T	VISHAY	VO2223A-X007T	VISHAY
120Ω	R1	R0603	1	0603WAF1200T5E	UNI-ROYAL	CRCW0603120RFKEA	VISHAY
100Ω	R3	RES-SMD_L6.4-W3.2-R2512	1	SR2512FK-7W100RL	YAGEO	100Ω SMD resistor	HTR-INDIA, Watts Electronics, Kusum/KOHM